

A PROJECT REPORT
ON
Machine Learning-Based Prediction of Organic Compound Adsorption onto
Microplastics in Freshwater

Submitted to



DES PUNE
UNIVERSITY

DES PUNE UNIVERSITY
SCHOOL OF SCIENCE AND MATHEMATICS
DEPARTMENT OF LIFE SCIENCES
M.Sc. BIOINFORMATICS

By

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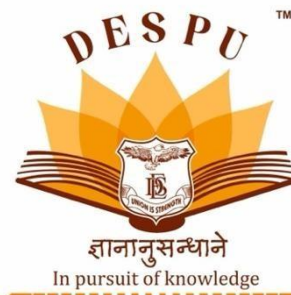
M.Sc. Bioinformatics (Semester III)

(Academic Year: 2026-2027)

As a part of '**Mini Project**' for the course
SMBI603: Machine Learning and Artificial Intelligence in Bioinformatics

Under the guidance of

Dr. Kushagra Kashyap



DES PUNE UNIVERSITY

SCHOOL OF SCIENCE AND MATHEMATICS

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SMBI603: Machine Learning and Artificial Intelligence in Bioinformatics

CERTIFICATE

This is to certify that **Harshada Risbud and Nikita Khandare** has satisfactorily completed the mini-project entitled “**Machine Learning-Based Prediction of Organic Compound Adsorption onto Microplastics in Freshwater**” as a part of internal assessment and curriculum of M.Sc. Bioinformatics, Semester-III during the academic year 2026-2027.

Place: Pune

Date:16/09/2026

Internal Guide

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DECLARATION

I affirm that the mini-project titled “**Machine Learning-Based Prediction of Organic Compound Adsorption onto Microplastics in Freshwater**” has been conducted by me under the guidance of Dr. Kushagra Kashyap. The findings presented in this report are authentic and represent the outcome of my work done at DES Pune University, Pune. None of the data presented here has been previously submitted to any other University or Institute for the award of any degree, and the data mentioned has been acquired through genuine work. Any relevant material taken from the open literature has been referred to and cited as per established ethical norms and practices. I also confirm that all the material which has been taken from other sources and incorporated in this project report is duly acknowledged. If any material is not duly acknowledged and is found incorporated in this report, it is entirely my responsibility. I am fully aware of the implications of any such act which might have been committed by me, advertently or inadvertently.

Place: Pune

Student's Name: Harshada Risbud
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Date:16/09/2026

ACKNOWLEDGEMENT

I would like to thank my guide, Dr. Kushagra Kashyap, for the SMBI603 course and for the guidance he gave me while this mini-project was taking shape. The idea of checking my model on unseen chemical structures rather than only on a random split came out of the way the subject was taught, and it ended up being the most useful part of the whole project.

I am thankful to Dr. Tushar Deshmukh, Head of the School of Science and Mathematics, and to the Department of Life Sciences, DES Pune University, for providing the facilities and the academic environment needed to carry out this work.

I also thank my classmate in M.Sc. Bioinformatics for the many discussions we had while debugging code and arguing about which evaluation numbers actually mean something. Finally, I thank my family for their constant support.

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ABSTRACT

Microplastics act as carriers for organic pollutants in freshwater, and the strength of this interaction is described by the sorption coefficient K_d . Measuring K_d experimentally is slow, so predictive models are useful. In this mini-project, a curated dataset of 1101 adsorption records covering 233 unique chemical structures was used to predict $\log K_d$ from 27 RDKit molecular descriptors, microplastic particle size, polymer category and water environment. Six regressors (a mean-value dummy, MLR, Ridge, Random Forest, SVM and XGBoost) were compared under two evaluation schemes: a conventional random-record 80/20 split and a chemical-structure-grouped split in which no compound appears in both training and test sets. XGBoost performed best in both schemes, but its R^2 fell from 0.864 on the random split to 0.658 on the grouped split, and grouped five-fold cross-validation gave a mean R^2 of 0.753 ± 0.019 . The random split therefore overstates performance, and the grouped result is the honest estimate of how the model would behave on a genuinely new compound.

1. BACKGROUND & INTRODUCTION

Plastic waste that breaks down in the environment eventually ends up as microplastics, which are conventionally defined as plastic particles smaller than 5 mm. These particles are now found in rivers, lakes, drinking water and sediments almost everywhere they have been looked for. What makes them interesting from an environmental chemistry point of view is not only the particles themselves, but what travels with them. Microplastics are hydrophobic and have a high surface-area-to-mass ratio, so organic pollutants already present in water tend to stick to them. Once that happens, the particle behaves as a carrier: a pollutant that would otherwise stay dissolved and dilute can be concentrated on a particle, transported somewhere else, and released again, or taken up by an organism that eats the particle.

The usual way to quantify how strongly a compound sticks to a given plastic is the sorption partition coefficient, K_d (L/kg), which is the ratio of the concentration of the compound on the plastic to its concentration in water at equilibrium. Because K_d spans several orders of magnitude, it is almost always reported on a log scale as $\log K_d$. A high $\log K_d$ means the compound prefers the plastic surface; a low value means it stays in water. Adsorption is driven by a combination of hydrophobic partitioning, van der Waals and dispersion forces, π - π stacking where aromatic rings are involved, hydrogen bonding and electrostatic interactions, and which of these dominates depends on both the compound and the polymer.

The problem is that K_d has to be measured one compound-polymer pair at a time, under one set of water conditions at a time. Each batch experiment takes days, and the number of possible combinations is enormous: tens of thousands of organic chemicals are in commercial use, and each can interact with polyethylene, polypropylene, polystyrene, PVC, PET and others, at different particle sizes and in different water chemistries. The published data therefore covers only a small corner of that space. This is exactly the situation where a predictive model is worth building — not to replace experiments, but to prioritise which experiments are worth doing.

Two families of computational approaches have been used for this. The older one is QSPR/QSAR, where a small number of carefully chosen molecular descriptors are put into a linear regression. Earlier work built such models for polyethylene, polypropylene and polystyrene and reported good fits, and later studies extended the idea to polyethylene and chlorinated polyethylene in freshwater. These models are interpretable, which is their main advantage, but they are narrow: each model applies to one polymer, and several of the

descriptors they rely on (such as experimental logKOW) have to be measured rather than calculated, so the model cannot be applied to a compound that lacks those measurements.

The newer approach uses general machine learning regressors on larger, more heterogeneous datasets. Other studies compared random forest, support vector machines and artificial neural networks for microplastic/water partition coefficients and obtained correlation coefficients above 0.92 in their query phase. Another assembled a literature dataset and used machine learning to predict aqueous sorption of organic pollutants on microplastics. Most recently, one study applied seven algorithms, including XGBoost, CatBoost and LightGBM, to 173 freshwater logKd values and used SHAP analysis to show that hydrophobicity, unsaturation and hydrogen-bond accepting ability were the main drivers of adsorption.

While reading these papers, one methodological point stood out. Adsorption datasets are built by pooling experiments from many publications, and the same compound usually appears several times, measured on different polymers, at different particle sizes, or by different groups. If such a dataset is split randomly into training and test sets, some records for a compound land in training and others in test. A tree-based model can then partly memorise the compound rather than learn a structure-property relationship, and the test score comes out optimistically high. The useful question is “given a new compound that has never been tested, how well can the model predict its logKd?” Answering it requires grouping the split by chemical structure so that no compound crosses the train-test boundary.

This mini-project is built around that comparison. The same dataset, the same features and the same six models are evaluated twice: once with a conventional random-record split, and once with a chemical-structure-grouped split. The gap between the two numbers is itself one of the results.

2. OBJECTIVES

1. To assemble and validate a curated dataset of experimental logKd values for organic compound adsorption onto microplastics, along with molecular and particle-level descriptors.
2. To build a reproducible Python pipeline for preprocessing that avoids test-set leakage.
3. To train and compare six regression models — dummy baseline, MLR, Ridge, Random Forest, SVM (RBF kernel), and XGBoost — for predicting logKd.
4. To evaluate each model under both a random-record split and a chemical-structure-grouped split, where no compound appears in both training and test sets.
5. To confirm result stability via five-fold cross-validation under both splitting schemes, reporting R^2 , RMSE and MAE.
6. To identify the key molecular and physical drivers of the best model's predictions using feature importances and SHAP values.
7. To quantify how much a random split overstates performance compared to a structure-aware split, and discuss the implications for adsorption models in the literature.

3. MATERIALS & METHODS

3.1 Overall workflow

The study was carried out as a single reproducible Python pipeline. The dataset was loaded and validated, the predictors and target were separated, two different train-test splitting strategies were applied, six models were trained under each strategy, and the results were confirmed by cross-validation before the best model was interpreted.

3.2 Dataset

The dataset (MPAP_Data.csv) contains 1101 experimental adsorption records described by 37 columns. The target variable is $\log K_d$, the base-10 logarithm of the microplastic/water partition coefficient. Each record also carries a canonical SMILES string for the organic compound and a Chemical_Group_ID, which is an identifier assigned so that all records describing the same chemical structure share the same value. There are 233 unique chemical structures across the 1101 records, i.e. on average each compound contributes about 4.7 measurements. It is this repetition that makes the grouped split necessary. A summary of the dataset is given in Table I.

Table I. Summary of the curated MPAP dataset.

Item	Value
Total records	1101
Total columns	37
Unique chemical structures (Chemical_Group_ID)	233
Average records per chemical structure	≈ 4.7
Target variable	$\log K_d$
Missing values in modelling columns	0
Numeric predictors	28
Categorical predictors	2

Before modelling, the dataset was validated programmatically. The pipeline raises an error and stops if the target, the group identifier or the SMILES column contains any missing value, if any nominally numeric descriptor fails to convert to a number, or if any categorical predictor is missing. No such error was raised, so the 1101 records used here are complete in every column that enters the model. Doing this as a hard check rather than a silent fillna was deliberate a descriptor that quietly becomes NaN and is then imputed is a bug that is very easy to miss.

3.3 Predictor variables

Twenty-seven molecular descriptors calculated with RDKit were used to represent the organic compound. They cover size and bulk (molecular weight, heavy atom count, molar refractivity, Labute approximate surface area, number of valence electrons), hydrophobicity and polarity (MolLogP, TPSA), hydrogen bonding (donor and acceptor counts), flexibility and shape (rotatable bonds, FractionCSP3, Kappa1-3), ring systems (total, aromatic, aliphatic and saturated ring counts), topology and complexity (BalabanJ, BertzCT, Chi0, Chi1), heteroatom content, and partial-charge extremes (maximum, minimum and absolute maximum/minimum Gasteiger partial charges).

One further numeric predictor, Average_size_um, describes the microplastic particle rather than the compound: smaller particles have more surface area per unit mass and generally adsorb more. Two categorical predictors were also included: Category, which encodes the chemical class of the organic compound, and Water_environment, which encodes the aqueous conditions of the experiment. The full list is given in Table II.

Table II. Predictor variables used in the models.

Group	Variables	n
Size / bulk	MolWt, MolMR, HeavyAtomCount, LabuteASA, NumValenceElectrons	5
Hydrophobicity / polarity	MolLogP, TPSA	2
Hydrogen bonding	NumHDonors, NumHAcceptors	2
Flexibility / shape	NumRotatableBonds, FractionCSP3, Kappa1, Kappa2, Kappa3	5
Ring systems	RingCount, NumAromaticRings, NumAliphaticRings, NumSaturatedRings	4
Topology / complexity	BalabanJ, BertzCT, Chi0, Chi1	4
Heteroatoms	NumHeteroatoms	1
Partial charges	MaxPartialCharge, MinPartialCharge, MaxAbsPartialCharge, MinAbsPartialCharge	4
Microplastic property	Average_size_um	1
Categorical	Category, Water_environment	2

3.4 Preprocessing

All preprocessing was placed inside a scikit-learn ColumnTransformer wrapped in a Pipeline, so that every transformation is fitted only on training data and then applied to test data. This matters: if scaling statistics or encoder categories are learned from the whole dataset, test

information leaks into training and the reported scores become meaningless. Numeric predictors were median-imputed and then standardised to zero mean and unit variance. Categorical predictors were imputed with the most frequent value and one-hot encoded with `handle_unknown` set to “ignore”, so that a category seen only in the test set does not crash the model. Standardisation is required by Ridge regression and by the RBF-kernel SVM; it is harmless for the tree models, which are scale-invariant, so applying it uniformly kept the pipeline simple.

3.5 Data splitting strategies

Two splitting strategies were used, and this is the methodological core of the project.

Random-record split. The 1101 records were split 80/20 at random with a fixed random state of 42, ignoring chemical identity. This is the conventional approach and is what most published studies report. Because a compound contributes several records, the same structure can appear on both sides of the split.

Chemical-structure-grouped split. `GroupShuffleSplit` was used with `Chemical_Group_ID` as the grouping variable, again with a 20% test fraction and random state 42. This guarantees that all records of a given compound stay together. The result was 876 training records covering 186 chemical structures and 225 test records covering 47 chemical structures, with zero overlap between the two sets of structures. The pipeline verifies this explicitly by intersecting the two sets of group IDs and raising an error if the intersection is non-empty. The overlap was zero, confirming no chemical leakage.

3.6 Models

Six regressors were trained under each splitting strategy. A `DummyRegressor` predicting the training mean was included as a floor: any model that cannot beat it has learned nothing. Multiple linear regression and Ridge regression ($\alpha = 1.0$) represent the classical QSPR-style linear approach, with Ridge adding L2 regularisation to cope with the strong correlations between molecular descriptors. Random Forest (500 trees, square-root feature subsampling) and XGBoost (500 trees, maximum depth 6, learning rate 0.05, subsample 0.8, column subsample 0.8) represent tree ensembles. A support vector regressor with an RBF kernel ($C = 10$, $\gamma = \text{scale}$, $\epsilon = 0.1$) represents kernel methods. Hyperparameters were set to reasonable literature defaults rather than tuned, because tuning on the test set would defeat the purpose of

the comparison, and the dataset is small enough that an extra nested tuning loop would have added noise rather than information. The settings are listed in Table III.

Table III. Model hyperparameters used in this study.

Model	Key settings
Dummy	strategy = mean
MLR	ordinary least squares, no regularisation
Ridge	alpha = 1.0
Random Forest	n_estimators = 500, max_features = sqrt, random_state = 42
SVM (SVR)	kernel = rbf, C = 10, gamma = scale, epsilon = 0.1
XGBoost	n_estimators = 500, max_depth = 6, learning_rate = 0.05, subsample = 0.8, colsample_bytree = 0.8

3.7 Evaluation metrics and cross-validation

Three metrics were reported: the coefficient of determination R^2 , the root mean squared error (RMSE) and the mean absolute error (MAE), all in $\log K_d$ units. RMSE penalises large errors more heavily than MAE, so the gap between them indicates how much of the total error comes from a few badly predicted compounds. Because a single 80/20 split of 1101 records can be lucky or unlucky, both schemes were repeated as five-fold cross-validation KFold with shuffling for the random scheme and GroupKFold on Chemical_Group_ID for the grouped scheme and the mean and standard deviation across folds were recorded.

3.8 Model interpretation

For the best-performing tree model on the grouped test set, built-in feature importances were extracted from the fitted estimator and the top 20 features were plotted. SHAP (SHapley Additive exPlanations) values were then computed with TreeExplainer on the transformed grouped test set, and features were ranked by mean absolute SHAP value. Feature importance tells you which variables the trees split on most profitably; SHAP tells you the direction and magnitude of each variable's contribution to individual predictions, so the two together are more informative than either alone.

3.9 Software

All work was done in Python 3 in a Jupyter Notebook on a Windows machine. The libraries used were pandas and NumPy for data handling, scikit-learn for preprocessing, the linear models, Random Forest, SVR, the dummy baseline, the splitters and the metrics, XGBoost for gradient boosting, SHAP for interpretation, Matplotlib for plotting and joblib for saving the

fitted pipelines. The molecular descriptors were calculated with RDKit. A fixed random state of 42 was used throughout so that the results can be reproduced exactly.

4. RESULTS & DISCUSSION

4.1 Performance on the random-record split

Table IV gives the test-set performance of the six models on the conventional random 80/20 split. XGBoost was the best model, with $R^2 = 0.864$, RMSE = 0.751 and MAE = 0.386 logK_d units, followed by Random Forest ($R^2 = 0.839$) and the SVM ($R^2 = 0.795$). The two linear models were clearly behind at $R^2 \approx 0.68$, and the dummy baseline sat at $R^2 = 0.000$ as expected.

Table IV. Test-set performance on the random-record 80/20 split (n = 220).

Model	R^2	RMSE	MAE
XGBoost	0.8644	0.7514	0.3864
Random Forest	0.8392	0.8183	0.4572
SVM	0.7953	0.9234	0.5385
MLR	0.6768	1.1602	0.8058
Ridge	0.6747	1.1641	0.8119
Dummy	-0.0000	2.0408	1.6962

Taken on its own, this table looks like a strong result and is comparable to what has been reported in the literature. It is also, as the next section shows, misleading.

4.2 Performance on the chemical-structure-grouped split

Table V gives the same six models evaluated on the grouped split, where the 47 test compounds are structures the models have never seen. Every model got worse, and the ranking changed. XGBoost was still best but dropped to $R^2 = 0.658$ (RMSE 0.937, MAE 0.671), a fall of 0.207 in R^2 and a 25% increase in RMSE relative to the random split. Random Forest fell from 0.839 to 0.608 and the SVM from 0.795 to 0.544.

Table V. Test-set performance on the chemical-structure-grouped 80/20 split (876 train / 225 test records; 186 / 47 structures; zero overlap).

Model	R^2	RMSE	MAE
XGBoost	0.6576	0.9374	0.6708
Random Forest	0.6084	1.0025	0.7277
SVM	0.5437	1.0821	0.7863
Ridge	0.4379	1.2011	0.9506
MLR	0.1793	1.4513	1.1417
Dummy	-0.1353	1.7069	1.3890

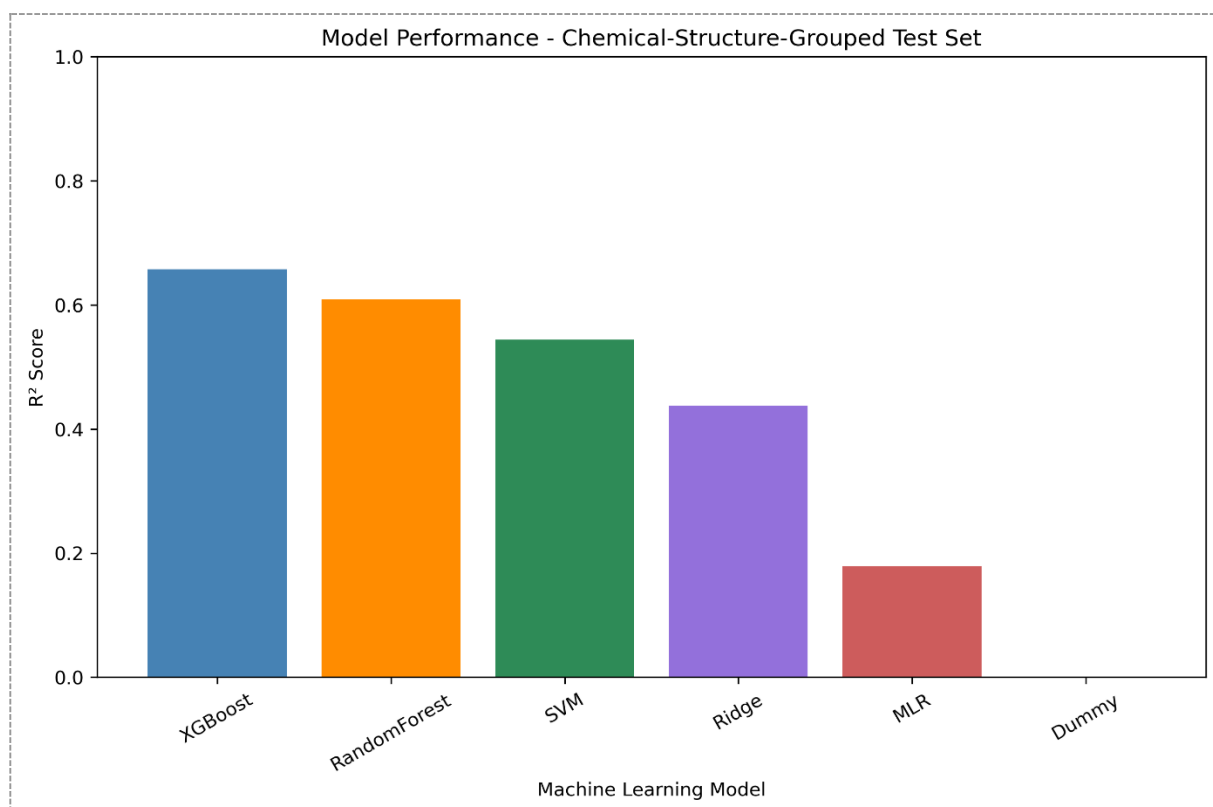


Figure 2. R² of the six models on the chemical-structure-grouped test set.

Two things in Table V are worth pausing on. First, the dummy baseline has a negative R² of -0.135 on the grouped split rather than the 0.000 it scores on the random split. This is not an error. R² is computed against the variance of the test set, and because the 47 test compounds were selected as whole structures, their logKd distribution is not the same as the training distribution. Predicting the training mean is therefore worse than predicting the test mean, and R² goes below zero. The negative baseline is itself evidence that the grouped test set is a genuinely different sample, which is exactly what it is supposed to be.

Second, MLR collapsed much harder than Ridge — 0.179 against 0.438. Both are linear models on the same features, so the difference is entirely due to L2 regularisation. The RDKit descriptors are heavily collinear (molecular weight, heavy atom count, molar refractivity and Labute ASA all measure size in slightly different ways), and unregularised least squares responds by fitting large coefficients of opposite sign that cancel each other out on the training data. Those coefficients extrapolate badly to unseen structures. Ridge shrinks them, and the improvement of 0.26 in R² is the cost of not regularising, made visible. On the random split the two models were indistinguishable (0.677 vs 0.675), so this failure mode is completely invisible unless the structure-aware split is done.

4.3 The gap between the two splits

Figure 3 compares XGBoost across the two strategies directly. The random split reports $R^2 = 0.864$; the grouped split reports 0.658. The difference of about 0.21 in R^2 is the amount by which a conventional evaluation would have overstated the model's ability to handle a new compound. In practical terms, the RMSE of 0.937 on the grouped split means that a prediction for a genuinely new chemical is typically off by roughly one $\log K_d$ unit, i.e. a factor of about ten in K_d itself. That is still useful for screening and prioritisation, but it is not the sub-0.8 error the random split advertises, and it should not be quoted as such.

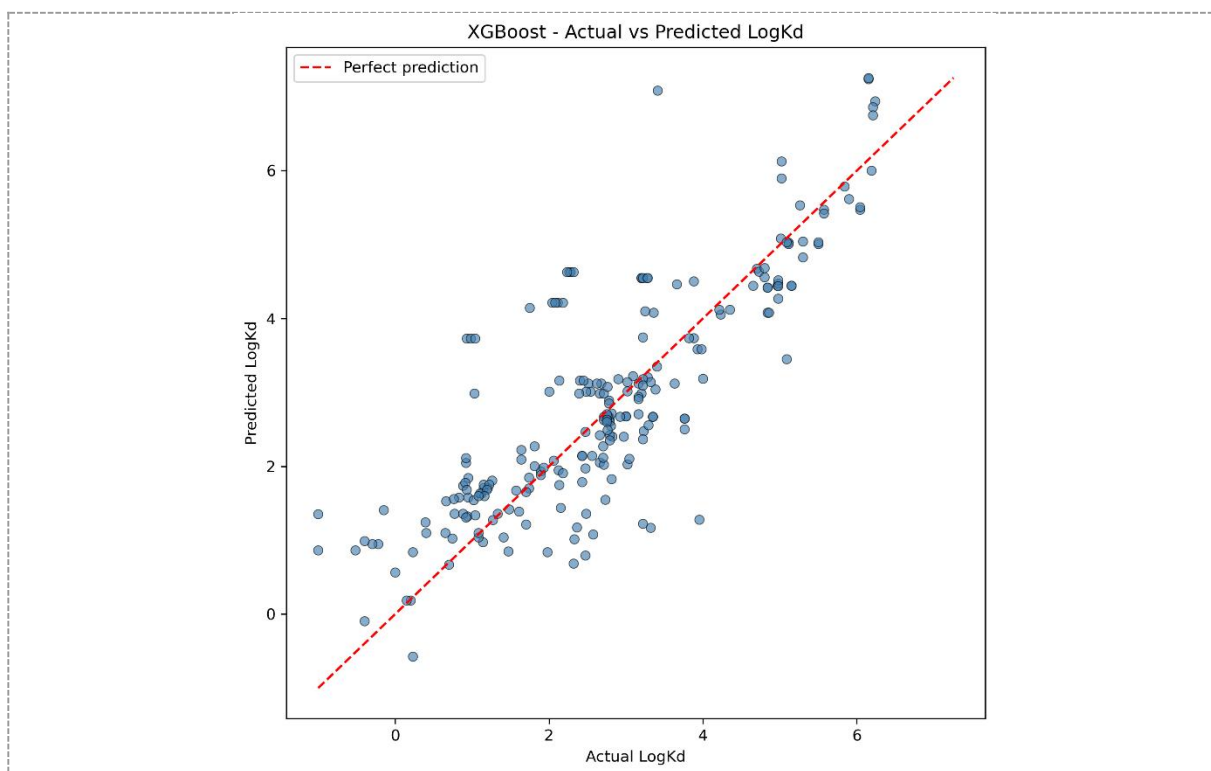


Figure 3. XGBoost R^2 under the random-record split and the chemical-structure-grouped split.

The mechanism is straightforward. When records of the same compound sit on both sides of a random split, a tree ensemble only has to recognise the descriptor fingerprint of that compound and recall the $\log K_d$ values associated with it, adjusting slightly for particle size or water environment. That is closer to a lookup than to a structure-property relationship. The grouped split removes the lookup and forces the model to generalise across chemical space, which is harder and which is the task that actually matters if the model is going to be used on an untested chemical.

4.4 Cross-validation

A single split can be lucky, so both schemes were repeated as five-fold cross-validation. Table VI gives the random five-fold results and Table VII the chemical-grouped results.

Table VI. Five-fold random cross-validation (mean $\pm$ SD across folds).

Model	Mean R ²	SD R ²	Mean RMSE	Mean MAE
XGBoost	0.885	0.037	0.660	0.389
Random Forest	0.864	0.023	0.726	0.444
SVM	0.847	0.030	0.770	0.487
MLR	0.700	0.023	1.077	0.801
Ridge	0.699	0.023	1.079	0.803
Dummy	-0.001	0.001	1.971	1.630

Table VII. Five-fold chemical-grouped cross-validation (GroupKFold on Chemical_Group_ID; mean $\pm$ SD across folds).

Model	Mean R ²	SD R ²	Mean RMSE	Mean MAE
XGBoost	0.753	0.019	0.955	0.692
Random Forest	0.733	0.035	0.991	0.718
SVM	0.687	0.056	1.069	0.757
Ridge	0.625	0.072	1.171	0.899
MLR	0.228	0.664	1.584	0.998
Dummy	-0.080	0.057	1.994	1.656

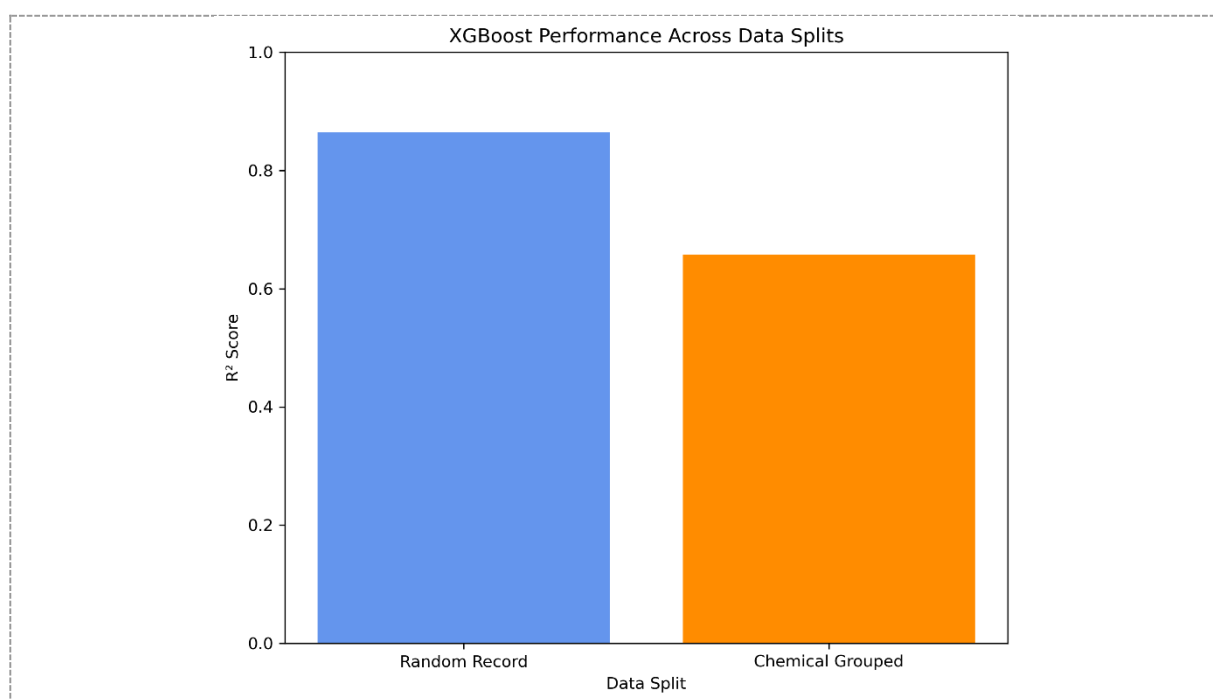


Figure 4. Mean R² from five-fold chemical-grouped cross-validation.

The cross-validated results confirm the single-split picture and add some detail. XGBoost is again best under both schemes, with a mean R^2 of 0.885 ± 0.037 randomly and 0.753 ± 0.019 when grouped. The grouped cross-validated value of 0.753 is higher than the 0.658 obtained on the single grouped split, which simply means that particular split drew a slightly awkward set of 47 test compounds; averaging over five folds is the more reliable estimate, and its small standard deviation of 0.019 says the estimate is stable.

The most striking row in Table VII is MLR, with a mean R^2 of 0.228 and a standard deviation of 0.664 — larger than the mean itself. This means that in at least one fold, unregularised linear regression produced predictions that were worse than useless, while in others it was reasonable. Ridge, on the same folds and the same features, gave 0.625 ± 0.072 . This is a clean demonstration that regularisation is not a cosmetic detail on collinear descriptor sets; it is what stops a linear model from occasionally falling apart when asked to extrapolate to new chemistry. It also suggests some caution about the older QSPR literature, where unregularised MLR on small datasets is common.

It is also worth noting that the ordering of models does not change between the two cross-validation schemes: XGBoost > Random Forest > SVM > Ridge > MLR > Dummy in both. So the grouped evaluation does not overturn the conclusion that gradient boosting is the right tool here; it changes how well we should expect that tool to perform.

4.5 Prediction quality of the best model

Figure 5 plots actual against predicted logKd for XGBoost on the grouped test set, and Figure 6 shows the corresponding residuals. Most points cluster around the 1:1 line, with scatter increasing towards the extremes of the logKd range. The residual plot shows no strong curvature, which suggests the model is not systematically mis-specified, but there is a visible tendency for the largest errors to occur at the high and low ends — the model is conservative, pulling extreme predictions towards the middle of the training distribution. This is normal for tree ensembles, which cannot extrapolate beyond the range of target values they were trained on: every prediction is an average of training leaf values, so the very strongest and very weakest adsorbers are always underpredicted and overpredicted respectively.

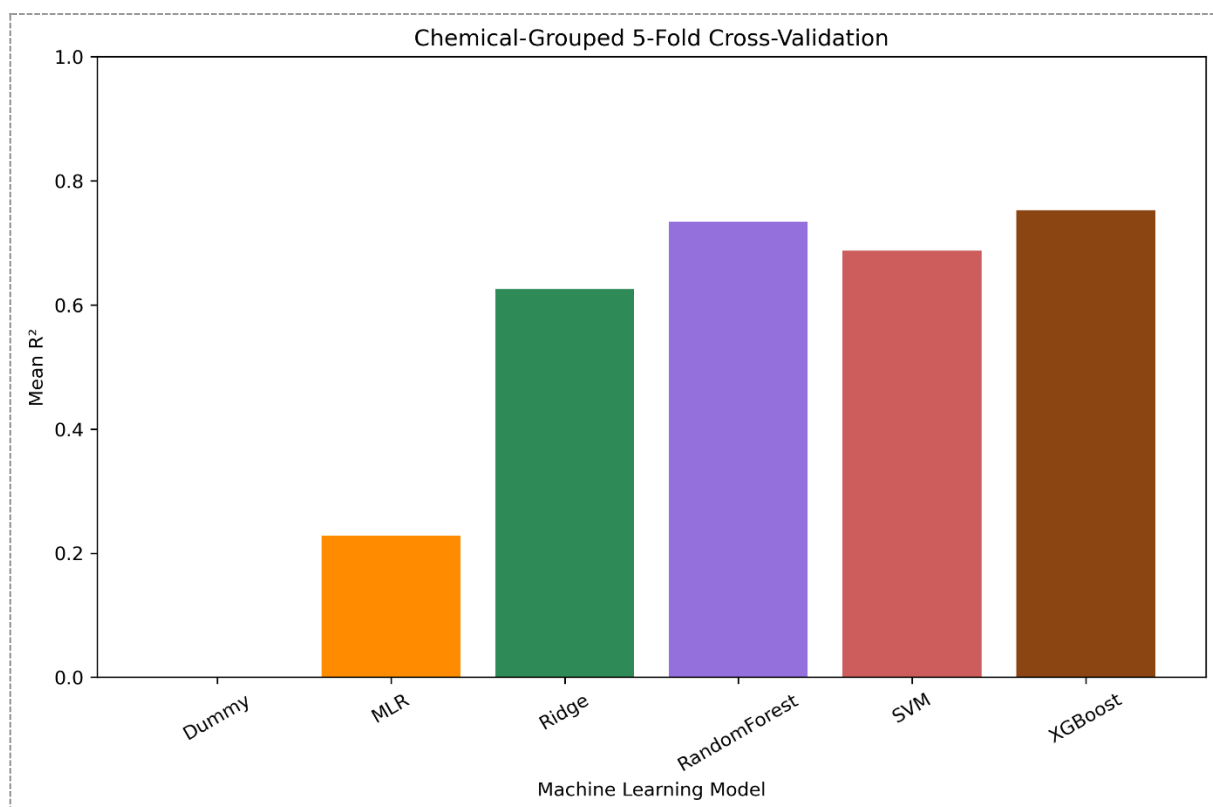


Figure 5. XGBoost actual vs predicted logKd on the chemical-grouped test set. The dashed line is the 1:1 line.

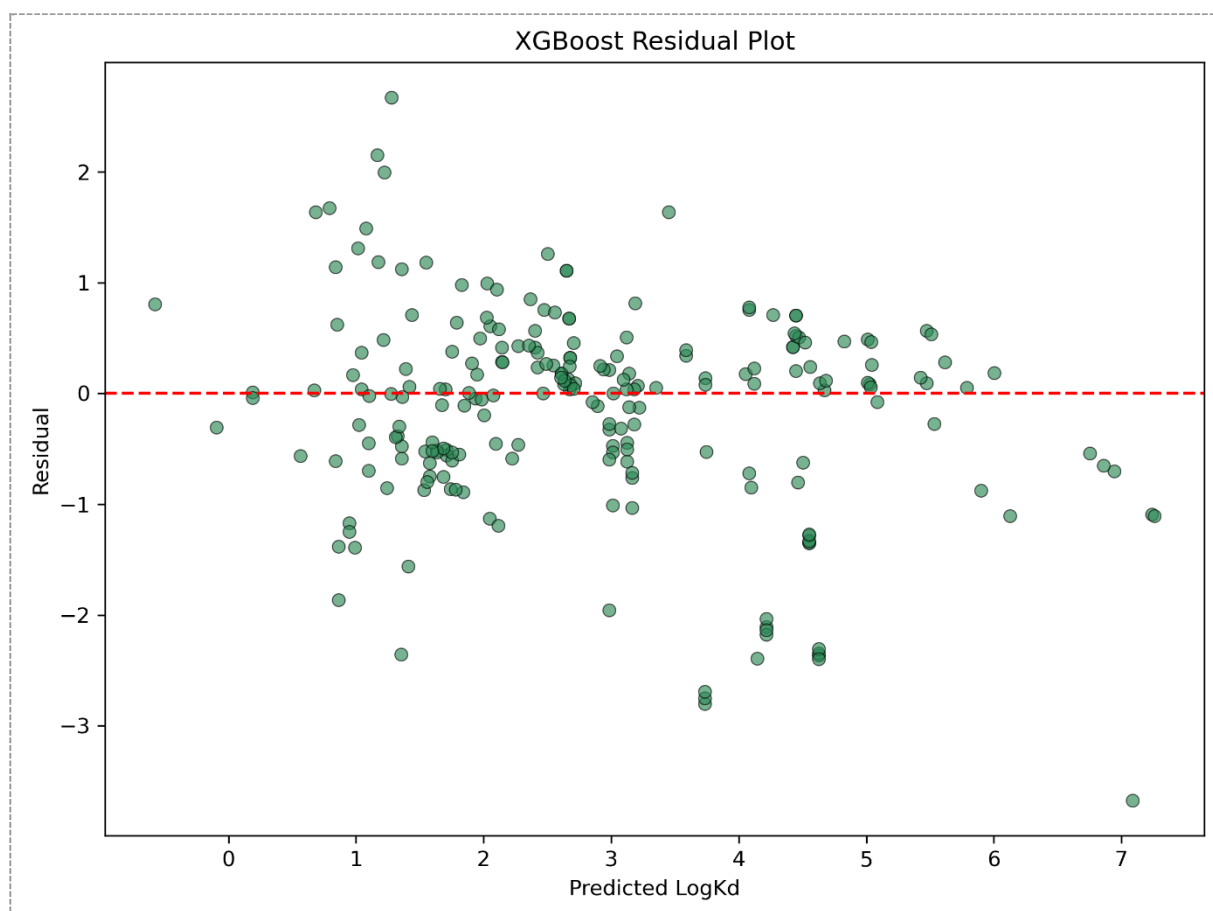


Figure 6. Residuals (actual – predicted) against predicted logKd for the XGBoost model.

The gap between RMSE (0.937) and MAE (0.671) on the grouped test set is also informative. If errors were uniformly distributed the two would be closer; the fact that RMSE is about 40% larger indicates that a minority of compounds are predicted much worse than the rest and dominate the squared error. Identifying those compounds and checking whether they share a structural feature would be a natural next step, and is noted in the future work section.

4.6 Feature importance and SHAP interpretation

XGBoost was the best model on the grouped test set, so it was interpreted using both built-in feature importances and SHAP values computed with TreeExplainer on the grouped test set. Figures 7 and 8 show the top 20 features by importance and the SHAP summary plot respectively.

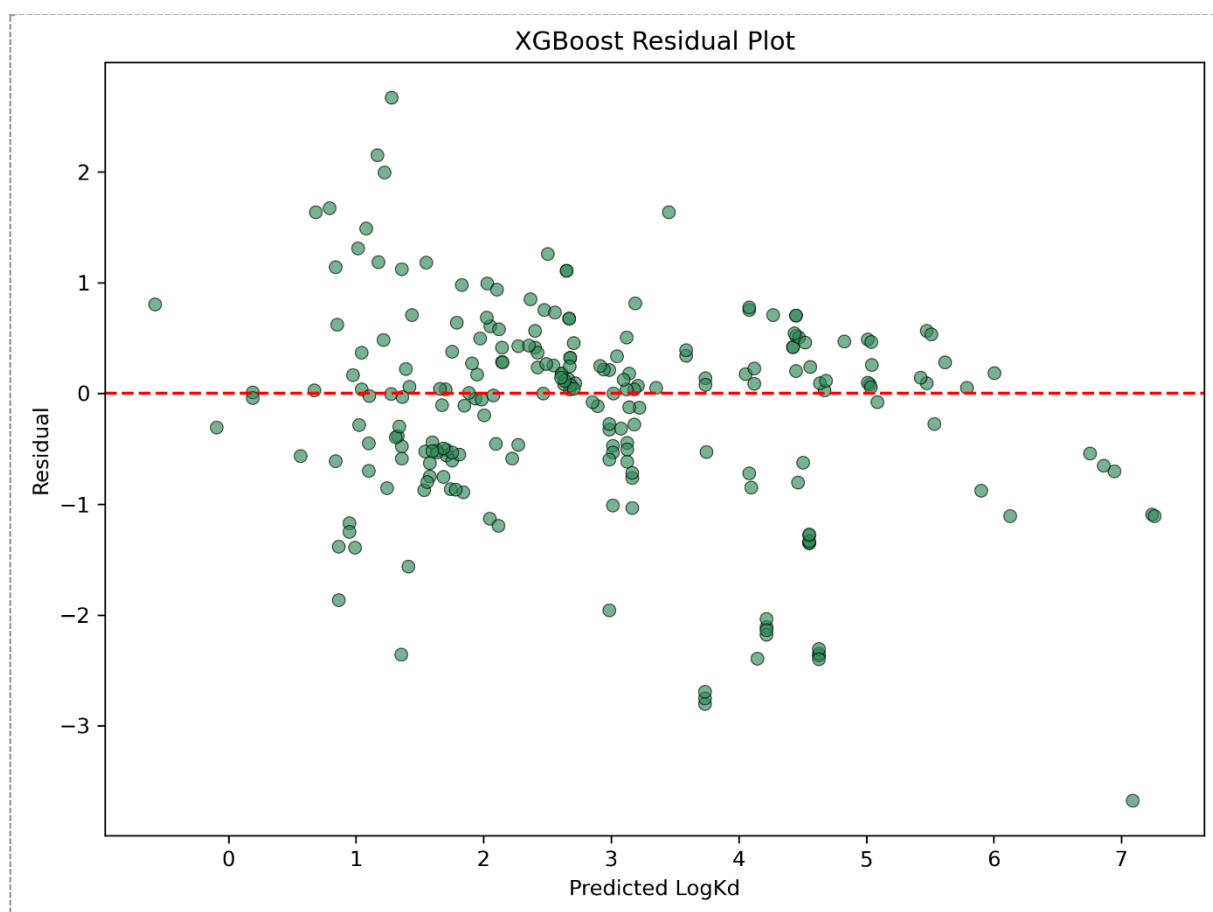


Figure 7. SHAP summary plot for the XGBoost model on the chemical-grouped test set.

[Fill in from your XGBoost_Feature_Importance.csv and SHAP_XGBoost_Importance.csv: name the top five features in each ranking and their values, then discuss them here. Expected chemistry, based on the literature, is that hydrophobicity-related descriptors (MolLogP, LabuteASA, MolMR, MolWt) should rank highly with a positive relationship to logKd, that TPSA and NumHAcceptors should contribute negatively because polar compounds prefer to

stay in water, that aromatic ring count should contribute positively through π - π interactions with the polymer, and that Average_size_um should contribute negatively because smaller particles present more surface area. State which of these your actual rankings support and which they do not — a mismatch is a result too, not a mistake.]

This part of the analysis connects the model back to chemistry. One recent study found through SHAP analysis that CrippenLogP was the most influential descriptor for most of their nonlinear models and that McGowan volume and unsaturation content correlated positively with logKd, while hydrogen-bond accepting ability correlated negatively. Another study likewise found partition-related variables dominant. If the descriptors that come out on top here follow the same pattern, it is reasonable evidence that the model has learned adsorption chemistry rather than dataset-specific artefacts — which matters more for a grouped-split model than for a random-split one, because the grouped model cannot have succeeded by memorisation.

4.7 Limitations

Three limitations should be stated plainly. First, the dataset is compiled from experiments done in different laboratories with different protocols, so part of the residual error is experimental scatter that no model can remove; the 0.94 logKd RMSE is therefore an upper bound on model error, not a pure measure of it. Second, hyperparameters were not tuned, so the numbers reported here are slightly pessimistic for every model, though the comparison between splits is unaffected because the same settings were used throughout. Third, 233 unique structures is a modest chemical space, and the applicability domain of the model has not been formally characterised — a prediction for a compound structurally far from the training set should not be trusted simply because the model returns a number.

5. CONCLUSIONS

Six regression models were compared for the prediction of $\log K_d$ for organic compounds adsorbing onto microplastics in freshwater, using 1101 records covering 233 unique chemical structures, 27 RDKit molecular descriptors, microplastic particle size and two categorical experimental variables. XGBoost was consistently the best model, followed by Random Forest, the SVM, Ridge and MLR, with a dummy baseline confirming that all five real models learned something.

The central finding is the size of the gap between the two evaluation strategies. On a conventional random-record split, XGBoost reached $R^2 = 0.864$ with an RMSE of 0.751. On a chemical-structure-grouped split, where no compound appears in both training and test sets, the same model reached $R^2 = 0.658$ with an RMSE of 0.937. Five-fold cross-validation gave the same picture more reliably: 0.885 ± 0.037 randomly versus 0.753 ± 0.019 grouped. Roughly 0.13 to 0.21 of R^2 in the conventional evaluation is therefore attributable to the same compounds appearing on both sides of the split, and not to genuine predictive ability on new chemistry.

A second finding is that regularisation matters far more than it appears to under a random split. MLR and Ridge were indistinguishable on the random split (R^2 0.677 vs 0.675) but separated sharply when grouped (0.179 vs 0.438), and in grouped cross-validation MLR was unstable to the point of having a standard deviation larger than its mean. On collinear molecular descriptor sets, unregularised linear regression should not be used for extrapolation to new compounds.

Practically, a grouped-split RMSE of about 0.94 $\log K_d$ units means predictions for untested compounds are typically within about one order of magnitude in K_d . That is accurate enough to rank compounds for experimental prioritisation or to flag likely strong adsorbers for risk assessment, but not accurate enough to substitute for measurement when a specific number is required. Reporting that honest figure, rather than the more flattering random-split figure, is the main contribution of this mini-project.

6. FUTURE DIRECTIONS

Several extensions follow naturally from this work.

1. Hyperparameter tuning within a nested grouped cross-validation loop, so that tuning is done without touching the outer test folds. This would give a fairer upper estimate of what gradient boosting can achieve on this dataset.
2. Formal characterisation of the applicability domain, for example using leverage values and Williams plots as in the QSPR literature, so that a prediction can be flagged as unreliable when the query compound is structurally far from the training set. Right now the model returns a number for any input, which is dangerous in a screening context.
3. Error analysis of the worst-predicted compounds in the grouped test set. The gap between RMSE and MAE suggests a minority of compounds dominate the error; checking whether they share a structural class, a polymer type or an experimental source would show whether the weakness is chemical or data-related.
4. Expanding the feature set with descriptors that encode the polymer explicitly — specific surface area, crystallinity, elemental ratios, degree of ageing — rather than treating polymer type as a bare category. Adsorption depends on both partners, and at present the compound side is described far more richly than the plastic side.
5. Extending the dataset beyond 233 structures, including seawater data alongside freshwater so the water environment variable has more to work with, and testing whether a model trained on freshwater transfers to seawater at all.
6. Trying molecular fingerprints or a graph neural network instead of fixed descriptors, and comparing them under the same grouped split. Learned representations sometimes generalise better across chemical space, but the comparison is only meaningful if the grouped evaluation is used for both.
7. Making the grouped-split evaluation routine rather than exceptional. If future adsorption modelling studies reported both numbers as a matter of course, the literature would be much easier to compare across.

7. REFERENCES

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